

Dylan Hammond

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EDUCATION

M.S. Mechanical Engineering: Utah State University (Robotics and Control Systems Emphasis)	Expected May 2026
B.S. Mechanical Engineering: Utah State University	Graduated May 2025
• Relevant Coursework: Mechatronics, Robotics Planning, Multivariable Control, Nonlinear Control, Deep Learning	
GPA: 3.77	

SKILLS

- **Programming:** Python, MATLAB, Simulink, C++, ROS2, Git
- **GNC & Robotics:** PID, LQR, Sensor Fusion, Embedded Systems
- **Tools & Software:** Abaqus, SolidWorks
- **Hardware & Manufacturing :** 3D Printing, Carbon Fiber Manufacturing, GD&T, Technical Drawings
- **Languages:** Portuguese (fluent)

PROJECTS

Autonomous Target-Tracking Turret	May 2025 – October 2025
• Designed autonomous pan-tilt turret using computer vision and servo actuation for real-time target tracking • Implemented closed-loop control system using PID controllers for stable tracking performance • Integrated OpenCV object detection with embedded motor control on Arduino	
Capstone Project: Hand-Signal Interpreter Glove	August 2024 – April 2025
• Collaborated with team of five to develop a wearable glove capable of interpreting 25 military hand signals at 95% accuracy • Programmed and calibrated IMU and flex sensors for real-time gesture data acquisition • Designed and deployed a 1D CNN, enabling embedded gesture classification on an ESP32 microcontroller • Integrated mode switching and RGB indicators for full-system control and feedback	
EKF Localization Robot	May 2026 - In Progress
• Developing ROS2-based differential drive robot for autonomous waypoint navigation using onboard sensing only • Implementing Extended Kalman Filter to fuse wheel encoder odometry and IMU data for real-time pose estimation • Validating filter performance in Gazebo simulation prior to deployment on physical hardware platform	
Aggie Maps	November 2025 – December 2025
• Developed a Python-based interactive path planning tool for pedestrian navigation on Utah State University campus • Implemented A* algorithm to compute optimal routes with dynamical cost functions for environmental constraints	

WORK EXPERIENCE

Product Development Intern	May 2025 – August 2025
Autoliv	Ogden, UT
• Optimized passenger airbag tether sizes to improve deployment safety through design validation and deployment testing • Designed a steel airbag deployment test fixture in SolidWorks for repeatable safety and failure-mode testing • Replicated passenger airbag tear failure modes through physical testing and analysis to support root-cause investigation	
Undergraduate Research Assistant	August 2024 – August 2025
EMAC Lab - Utah State University	Logan, UT
• Developed FE models in Abaqus to simulate electromechanical loading on materials for roadway electrification • Automated simulation post-processing using Python, reducing analysis time • Recognized as 2024 Undergraduate Scholar and winner of USU Undergraduate Poster Competition	
Photovoltaics and Materials Internship	May 2022 – August 2022
Sandia National Laboratories	Albuquerque, NM
• Optimized microgrid solar array power output through positioning and axis tracking in Python • Authored Sandia Labs technical paper on solar microgrid implementation strategies for small communities • Presented work and research to the National Renewable Energy Laboratory	